Impact & sustainability

National 5 · Impact · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe how design and manufacturing technologies have changed the workforce
- ☐ describe the impact on the supply of products and society
- ☐ describe the impact on energy use and pollution
- ☐ describe methods to support sustainability (6 Rs)

I DO — Worked example

Read how a full-mark answer is built

1 Describe four steps designers could take to make products more sustainable. (4 marks)

Full-mark answer each ✓ is one mark

Label each plastic part with its recycling code so it can be sorted ✓. Use screws rather than glue so the product can be taken apart at the end of its life ✓. Make parts such as batteries replaceable so the product lasts longer ✓. Use less material and less packaging ✓.

Why marks are not awarded

"Use recycled materials and make it in a factory with solar panels."

0–1 mark: in some years recycled materials were given in the question, and points about the factory do not answer "designers".

WE DO — Guided practice

Use the starters and word bank to help you

2 Describe the impact of mass manufacturing on the workforce. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

CAD · traditional · replace · jobs · needed · machines · workers · manual

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which design decision best supports recycling?
☐ Using one material and labelling it ☐ Gluing different plastics together ☐ Painting every part ☐ Using permanent fixings

b) Designing a product to be fixed easily, extending its life, is which R?
☐ Reduce ☐ Refuse ☐ Rethink ☐ Repair

c) Which is a negative impact of automation on the workforce?
☐ More jobs for CAD technicians and engineers ☐ Safer working conditions on the factory floor ☐ Loss of traditional craft skills and fewer jobs ☐ Higher accuracy and fewer faulty products

d) Sourcing materials locally mainly reduces...
☐ Product aesthetics ☐ Transport pollution ☐ Strength ☐ Branding

4 Key terms: write the letter of the matching description next to each term.

1. **Sustainability**	____	**A** Sustainable/local materials, less transport, less energy, less waste.
2. **The 6 Rs**	____	**B** Reduce, Rethink, Refuse, Recycle, Reuse, Repair.
3. **Clean manufacturing**	____	**C** Using resources responsibly so future generations can still meet their needs.

5 Describe the impact of automation on workforce. (2 marks)

6 Explain what is meant by ‘Circular economy’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?